QP CODE:	**25901614**
MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026	
SECOND SEMESTER - Discipline Specific Core Course (DSC) – MG2DSCBOT100 - PLANTRESOURCES AND VENTURES IN BOTANY	
(2024 ADMISSION ONWARDS)	
Duration: 1 Hour 30 Minutes	Max. Marks : 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A - Short Answer - Answer any **10** questions - Each question carries **1** marks

1	Write any two medicinal uses of Eucalyptus oil.	[U] / [CO1]
	Soothing Joint and Muscle Pain: Relieving Respiratory Congestion	
2	What part of Cotton plant is used for cotton fiber manufacture?	[K] / [CO1]
	Cotton fiber is manufactured from the seed hairs found in the fruit (testal outgrowths) of the cotton plant.	
3	What is the common use of bamboo in construction?	[K] / [CO1]
	Bamboo is commonly used in construction for structural framing, scaffolding, flooring, roofing, and wall panels ***(any one use)***	
4	What are soups?	[U] / [CO1]
	Soup is a primarily liquid, savory food made by simmering ingredients like meat, fish, or vegetables in stock, milk, or water.	
5	What is an Arboretum?	[K] / [CO5]
	An arboretum is a specialized botanical garden or, a "living museum" dedicated to the cultivation, study, and conservation of various species of trees and woody plants.	
6	Define biodiversity.	[U] / [CO2]
	Biodiversity, or biological diversity, is the immense variety of all living things on Earth including plants, animals, fungi, and microorganisms -along with the genetic differences within them and the complex ecosystems they inhabit	
7	Where is the headquarters of BSI located?	[K] / [CO2]
	The headquarters of the Botanical Survey of India (BSI) is located in Kolkata, West Bengal.	
8	Name any two artificial means of propagation.	[U] / [CO3]
	Budding, grafting, layering, cutting, micropropagation (any two)	
9	Name any two biofertilizer used in organic farming.	[K] / [CO3]

			Rhizobium: A symbiotic bacterium used for legume crops to fix nitrogen. Azotobacter: A free-living nitrogen-fixing bacterium widely used for cereals and vegetables.
10	Expand SBI. [K] / [CO5]		
			State Bank of India
11	What is blockchain technology? [K] / [CO4]		
			Blockchain technology is a decentralized, distributed, and immutable digital ledger used to record transactions across a network of computers.
12	Who is a Scientific assistant? [K] / [CO5]		
			A Scientific Assistant is a technical professional who provides hands-on support to scientists, researchers, or engineers by conducting experiments, collecting data, and maintaining laboratory equipment.
			(10×1=10)

Part B - Short Essay - Answer any 6 questions
Each question carries 5 marks

13	Describe the nutritional value and uses of rice? [K] / [CO1]		
			Nutritional Value of Rice ❖ Carbohydrates: Rice is a dense source of carbohydrates, providing energy to the body. ❖ Protein and Amino Acids: It contains 7–8% protein and is a good source of lysine, an amino acid that helps in wound healing and reducing anxiety. ❖ Fiber: Brown, red, and black rice contain high amounts of fiber, aiding digestion. ❖ Vitamins and Minerals: Rice is rich in manganese, niacin thiamin, selenium, and iron. ❖ Antioxidants: Bran/germ (especially in brown/pigmented rice) contains ferulic acid and lignans, which have anti-inflammatory and cancer-preventive properties. Comparison: Whole grain (brown) rice retains the bran and germ, making it more nutritious than white rice, which is polished and lacks these layers.
14	How does plant based bio manure improve soil health ? [U] / [CO1]		
			❖ Improves Soil Structure and Aeration: Organic manure binds soil particles together, improving texture, enhancing aeration, and promoting better root penetration ❖ .Increases Water-Holding Capacity: By adding organic matter, the soil can retain more water, reducing the need for frequent irrigation and protecting plants against drought. ❖ Boosts Microbial Activity: It serves as a food source for beneficial soil bacteria, fungi, and earthworms, fostering a healthy microbiome that aids nutrient cycling. ❖ Provides Slow-Release Nutrients: Unlike chemical fertilizers, it gradually releases nutrients like nitrogen, phosphorus, and potassium (NPK), providing sustainable nourishment to crops without toxicity. ❖ Increases Soil Organic Carbon: It increases the soil organic carbon content, which acts as a carbon sink and directly improves soil health indicators. ❖ Suppresses Diseases and Pests: Many plant-based manures contain organic compounds that have antimicrobial properties, helping to suppress soil-borne diseases.

15	Briefly explain the role of trees in reducing urban air pollution.	[U] / [CO1]
	Key Mechanisms: ➢ Pollutant Absorption: Leaves absorb gaseous toxins through tiny pores called stomata ➢ .Particulate Filtration: Leaves and bark trap dust and smoke particles which are then washed to the ground by rain. ➢ Carbon Sequestration: Trees absorb carbon dioxide and store it, reducing greenhouse gases. ➢ Temperature Reduction: Through shading and transpiration, trees cool urban areas, reducing the formation of ground-level ozone. ➢ Reduced Energy Usage: By shading buildings, trees lessen the need for air conditioning, indirectly lowering emissions from power plants.	
16	Write a note on copyrights.	[U] / [CO4]
	Copyright is a form of intellectual property law that protects original works of authorship as soon as they are fixed in a tangible form of expression. It grants the creator exclusive rights to control how their work is used, reproduced, and distributed. Major Aspects ❖ Automatic Protection: Copyright exists the moment a work is created and fixed (e.g., written down, recorded, saved to a disk). ❖ What is Protected: Original works, including literary (books, poems), artistic (paintings, photographs), musical compositions, sound recordings, movies, software, and architectural designs. ❖ What is NOT Protected: Ideas, procedures, methods of operation, or mathematical concepts are not protected, only the expression of those ideas ❖ .Duration: varies with country ❖ Rights Granted: Owners have the exclusive right to reproduce, distribute, perform, display, and create derivative works.	
17	What is layering? Name any two types of layering.	[K] / [CO3]
	Layering is a vegetative method of plant propagation where a stem or branch is encouraged to develop roots while still attached to the parent plant. Once roots form, the new plant is detached and grown independently. Layering is a highly successful, low-maintenance, and reliable method of vegetative propagation that allows new plants to develop roots while still receiving water, nutrients, and hormones from the parent plant. It is ideal for woody plants and ensures high success rates by creating a large, mature, and genetically identical, new plant. Two common types of layering are: ❖ Air Layering (Marcottage): A section of an aerial stem is wounded, wrapped with moist moss/soil, and covered with plastic to induce rooting. ❖ Simple Layering (Ground Layering): A low-growing branch is bent to the ground, partially buried, and anchored to form roots.	
18	Give a brief account on CTCRI.	[U] / [CO2]

		The ICAR-Central Tuber Crops Research Institute (CTCRI) is the world's only research organization exclusively dedicated to tropical tuber crops (cassava, yams, taro, sweet potato, etc.). Established in 1963 in Thiruvananthapuram, Kerala, it functions under the Indian Council of Agricultural Research (ICAR) to enhance productivity, sustainability, and utilization of these crops for food security. ❖ Headquarters: Sreekariyam, Thiruvananthapuram, Kerala.Regional Station: A regional center exists in Bhubaneswar, Odisha, ❖ .Achievements: The institute has released over 77 improved varieties and developed numerous post-harvest technologies, including bio-plastics and industrial starch applications. ❖ Mandate: Focuses on improving tuber crops to ensure food security, nutrition, and livelihood for rural, resource-poor farmers.
19	Give an account on market assessment in entrepreneurship. [K] / [CO4]	
		Market assessment in entrepreneurship is the systematic evaluation of a specific market's potential, attractiveness, and dynamics to determine if a business idea is viable. It enables entrepreneurs to understand customer needs, analyze competition, estimate demand, and reduce risks before investing resources. This process combines quantitative data (size, growth) and qualitative insights (consumer preferences, trends) ❖ Components of a market assessment - Market Dynamics, .Customernalysis:.Competitive Landscape: Regulatory & Environmental Factors etc ❖ Methods of Market Assessment - -:Primary Research: &Secondary Research ❖ Significance for Entrepreneurs - Risk Mitigation. Strategic Planning:. Validation & Funding:

20	Distinguish between Horticulturist and Taxonomist. [U] / [CO5]			
			Horticulturist	Taxonomist
		Definition:	A professional who studies and practices the cultivation, management, and improvement of fruits, vegetables, flowers, and ornamental plants.	A scientist who specializes in identifying, naming, describing, and classifying plant species based on their evolutionary relationships and characteristics.
		Goal:	To improve plant yield, quality, and aesthetic value.	To understand plant diversity and organize plants into a formal, orderly system
		Key Activities:	Plant breeding, greenhouses/nursery management, landscape design, and plant care.	Identifying new plant species, organizing herbariums, and determining phylogenetic relationships.
		Work Environment	Plant breeding, greenhouses/nursery management, landscape design, and plant care.	Frequently in laboratories, offices, or performing field research in diverse ecosystems.
		Key Terms:	Cultivar, propagation, Pomology, Olericulture, Floriculture, Landscape	Genus, species, Phylogeny, Morphology, Nomenclature, Herbarium.
		(***any relevant three point -full credit, liberal valuation)***		
				(6×5=30)

Part C – Essay - Answer any 1questions Each question carries 10 marks			
21	Explain the economic potential of plant based enterprises. Illustrate citing examples of 5 plants used in Cosmetics, Skin Care and Hair Care. Outline their Binomial, Family, Part used and Specific uses.		[U] / [CO1]
		The economic potential of plant-based enterprises is massive and rapidly expanding, driven by consumer demand for sustainability, health, and ethical consumption. This sector is restructuring food systems by creating new jobs, increasing agricultural profitability, and diversifying food supply chains. Key aspects of the economic potential include: a) High Market Growth and Investment -Rapid Expansion and segment strength b) Job Creation and Economic Contribution – job growth, high earning, GDP and taxes etc. c) Agricultural and Farming Opportunities - Diversification and Profitability:, lower production costs , sustainable agriculture etc d) Innovation and Technology- novel products, technological advancement and export potential	
		Plants used in cosmetics: ❖ Cucumber (Cucumis sativus) - Cucumbers contain natural astringent properties that help tighten pores, control sebum, and cleanse oily or acne-prone skin. The presence of organic acids like glycolic and lactic acid acts as a gentle exfoliant, removing dead skin cells.. ❖ Rose: Renowned for high vitamin C content, rose water acts as a gentle astringent, skin toner, and fragrance agent ❖ ..Neem (Azadirachta indica): Valued in Ayurvedic cosmetics for its powerful anti-inflammatory and anti-acne properties, often found in soaps and face washes	
		Plants used for skin care ❖ White Sandalwood (Santalum album): The most precious, aromatic type used for oil, incense, and skin treatments. ❖ Red Sandalwood (Pterocarpus santalinus): A different species in the Fabaceae family, used as a mild exfoliant and skin brightener rather than for its scent. ❖ Turmeric (Curcuma longa): Contains curcumin, which helps brighten the skin, reduce acne, calm inflammation, and even out skin tone.	
		Plants used for hair care ❖ .Bhringraj (Eclipta alba): Often called the "king of herbs" for hair, it is used to strengthen roots, promote hair growth, and combat dandruff. It is commonly applied as an oil (Bhringraj oil). ❖ Indian Gooseberry (Amla): Rich in Vitamin C and antioxidants, Amla strengthens hair, promotes growth, and supports scalp hygiene. It is commonly used in powdered form or as an infused oil ❖ Henna (Lawsonia inermis) is a natural, plant-based hair care solution derived from an evergreen shrub that offers deep conditioning, strengthens hair follicles, fights dandruff, and provides a reddish-orange dye through the molecule lawsone ***(Economicl potential with atleast 4 points - 4 marks, |Any two plants with binomial and ,morphology from each category = 6 marksy***	

22	What are the applications challenges, concerns and future prospects of Genetic Engineering?		
		Genetic engineering (GE) involves altering an organism's DNA using technology to create desired traits, with major applications in medicine (gene therapy, vaccines), agriculture (pest-resistant, biofortified crops), and industry. While promising huge benefits, it faces technical challenges like off-target mutations, significant environmental concerns, and ethical dilemmas regarding designer babies	
		Applications of Genetic Engineering Medicine: Gene therapy to correct genetic disorders, production of insulin and vaccines, and cancer treatment (e.g., CAR-T cell therapy).Agriculture: Development of genetically modified (GM) crops with enhanced resistance to pests, diseases, and environmental stresses (e.g., drought-resistant crops), alongside improving nutritional value. Industry & Environment: Engineered microorganisms used to produce industrial enzymes, biofuels, and bioremediation (cleaning up toxic waste).Research: Creating animal models to understand human diseases and gene function	
		Challenges Technical Limitations: Risks of off-target effects, where unintended parts of the genome are edited, leading to potential mutations .Environmental Impact: Potential for reduced biodiversity, gene flow to wild species, and the creation of invasive species. Economic/Regulatory: High development costs, complex regulatory approvals, and debates over patenting living organisms.	
		ConcernsEthical & Social: Concerns about "designer babies" (enhancing traits rather than curing diseases), genetic discrimination, and altering the human germline (inheritable changes).Safety & Health: Long-term effects of consuming GM foods are still debated, although scientific consensus suggests they are safe .Biosecurity: The potential to misuse technology to create harmful pathogens.	
		Future ProspectsCRISPR-Cas9 Advancements: Continued refinement of gene-editing tools for greater precision and efficiency. Sustainable Agriculture: Developing climate-resilient crops and reducing reliance on chemical pesticides. Personalized Medicine: Tailoring treatments to an individual's genetic makeup. Synthetic Biology: Creating entirely new biological parts and systems for innovative applications	
			(10×1=10)